PART 1: Augmented Lagrangian (2 Examples)

✅ Example 1 (Easy): Equality-constrained quadratic

$$\min_x \frac{1}{2}x^2 \text{s.t. } x = 1$$

Step 1: Augmented Lagrangian

$$\mathcal{L}_\rho(x, \lambda) = \frac{1}{2}x^2 + \lambda(x-1) + \frac{\rho}{2}(x-1)^2$$

Step 2: x-update

Differentiate:

$$x + \lambda + \rho(x-1) = 0$$
$$(1+\rho)x = \rho - \lambda$$
$$x^{k+1} = \frac{\rho - \lambda^k}{1+\rho}$$

Step 3: Dual update

$$\lambda^{k+1} = \lambda^k + \rho(x^{k+1} - 1)$$

Step 4: Numerical iteration

Let $\rho = 1, \lambda^0 = 0$

$$x^1 = \frac{1}{2}, \lambda^1 = 0 + (0.5 - 1) = -0.5$$
$$x^2 = \frac{1 - (-0.5)}{2} = 0.75$$

👉 Converges to $x = 1$

💡 **Insight**

- Penalty term pulls toward feasibility
- multiplier corrects bias

✅ Example 2 (Medium): Linear regression with constraint

$$\min_x (x-3)^2 \text{s.t. } x = 2$$

Augmented Lagrangian

$$\mathcal{L}_\rho = (x-3)^2 + \lambda(x-2) + \frac{\rho}{2}(x-2)^2$$

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x-update

Differentiate:

$$2(x-3)+\lambda+\rho(x-2)=0$$
$$(2+\rho)x=6+2\rho-\lambda$$
$$x^{k+1}=\frac{6+2\rho-\lambda^k}{2+\rho}$$

Dual update

$$\lambda^{k+1}=\lambda^k+\rho(x^{k+1}-2)$$

💡 **Insight**

- Objective wants $x=3$
- constraint forces $x=2$
- solution = constrained optimum

PART 2: ADMM (General Form)

✅ **Example 3 (Easy): L1 regularization (Lasso-like)**

$$\min_x \frac{1}{2}(x-4)^2+|x|$$

Step 1: Split

$$\min_{x,z} \frac{1}{2}(x-4)^2+|z| \text{ s.t. } x=z$$

Step 2: ADMM updates

x-update

$$x^{k+1}=\frac{4+\rho(z^k-u^k)}{1+\rho}$$

z-update (soft thresholding)

$$z^{k+1}=S_{1/\rho}(x^{k+1}+u^k)$$

dual

$$u^{k+1}=u^k+(x^{k+1}-z^{k+1})$$

Step 3: Numerical

Let $\rho=1, x^0=z^0=u^0=0$

$$x^1=2, z^1=1, u^1=1$$

💡 **Insight**

- x-step → smooth part

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- z-step → sparsity

✅ **Example 4 (Medium): Feature selection (2D)**

$$\min_x \frac{1}{2} \| x - b \|^2 + \lambda \| x \|_1$$

Let $b = [3, -1], \lambda = 1$

ADMM

$$x^{k+1} = \frac{b + \rho(z^k - u^k)}{1 + \rho}$$
$$z^{k+1} = S_{\lambda/\rho}(x^{k+1} + u^k)$$

Iteration

Start $x^0 = 0$

$$x^1 = [1.5, -0.5]$$

Apply soft threshold:

$$z^1 = [0.5, 0]$$

💡 **Insight**

- second feature shrinks to 0
- shows **feature selection**

PART 3: ADMM for LP

✅ **Example 5 (Easy)**

$$\min_x \ x \quad \text{s.t. } x \geq 0$$

Split

$$\min_{x,z} \ x + I_{z \geq 0}(z), x = z$$

Updates

$$x^{k+1} = z^k - u^k - \frac{1}{\rho}$$
$$z^{k+1} = \max(0, x^{k+1} + u^k)$$
$$u^{k+1} = u^k + (x^{k+1} - z^{k+1})$$

Iteration ($\rho = 1$)

$$x^1 = -1, z^1 = 0, u^1 = -1$$

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💡 **Insight**

- projection enforces feasibility

✅ **Example 6 (Medium)**

$$\min_x \ 2x \quad \text{s.t. } x \geq 1$$

Reformulation

$$z \geq 1$$

z-update

$$z = \max(1, x + u)$$

Iteration

$$x^1 = -2, z^1 = 1$$

💡 **Insight**

- projection onto shifted set

PART 4: ADMM for QP

✅ **Example 7 (Easy)**

$$\min_x \frac{1}{2}x^2 \text{s.t. } x \geq 0$$

x-update

$$(1 + \rho)x = \rho(z - u)$$
$$x^{k+1} = \frac{\rho(z^k - u^k)}{1 + \rho}$$

z-update

$$z = \max(0, x + u)$$

Iteration

Start 0:

$$x^1 = 0,\ z^1 = 0$$

✅ **Example 8 (Medium, AI relevant)**

$$\min_x \frac{1}{2}x^2 - 2x \text{s.t. } x \geq 0$$

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x-update

$$(1+\rho)x = 2 + \rho(z-u)$$
$$x^{k+1} = \frac{2+\rho(z^k-u^k)}{1+\rho}$$

Iteration

$$x^1 = 1, z^1 = 1, u^1 = 0$$

💡 **Insight**

- unconstrained optimum = 2
- constraint satisfied → stays feasible

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